

The empirical recurrence for OEIS A283723: recovering a conjecture that is not written down

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Abstract

OEIS A283723 records “Empirical recurrence of order 80”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 1370$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 80, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 80 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A283723 is “Number of $n \times 5$ 0..1 arrays with no 1 equal to more than two of its horizontal, diagonal and antidiagonal neighbors, with the exception of exactly one element.”. It has offset 1 and begins

0, 242, 7839, 185564, 4096541, 85216204, 1712274593, 33562500568, ...

The entry states:

Empirical recurrence of order 80 (see link above).

As of the “Last modified” line on the live entry (Jun 23 2026 13:04:42, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1370$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 1370$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 2740$ exact terms of the model returns order 80 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{80} c_i a(n-i),$$

c_1	28	c_{21}	−18516594505092	c_{41}	−11029040155348844	c_{61}	−424229788070
c_2	−138	c_{22}	−50167898439979	c_{42}	−6260343077863472	c_{62}	−25995514093
c_3	−252	c_{23}	−67970269167052	c_{43}	−726259091095298	c_{63}	141641933286
c_4	−6949	c_{24}	−55867361187449	c_{44}	1607873657720669	c_{64}	46434014311
c_5	−25368	c_{25}	45583681912784	c_{45}	1095163796047960	c_{65}	25385233858
c_6	−370740	c_{26}	291870757853588	c_{46}	−202941865936307	c_{66}	14383370138
c_7	−705080	c_{27}	744332070957264	c_{47}	−717760380132418	c_{67}	3870405652
c_8	−1958246	c_{28}	1002588098349287	c_{48}	−433042131648751	c_{68}	−301591613
c_9	12240056	c_{29}	551334152246572	c_{49}	101163437362170	c_{69}	−3293483296
c_{10}	51915542	c_{30}	−926031004446334	c_{50}	330703116603053	c_{70}	−545639956
c_{11}	582800130	c_{31}	−2694266434894652	c_{51}	215859317476766	c_{71}	−48977000
c_{12}	1736324874	c_{32}	−4211567445386086	c_{52}	55035148305188	c_{72}	154573052
c_{13}	1140589938	c_{33}	−4729152448609770	c_{53}	−31996212096338	c_{73}	−49554944
c_{14}	−29363937824	c_{34}	−2716545924895571	c_{54}	−33167581503763	c_{74}	6522864
c_{15}	−116349365388	c_{35}	3429394460406050	c_{55}	−10045398915972	c_{75}	12090912
c_{16}	−171520935311	c_{36}	11242681001388980	c_{56}	7069209610834	c_{76}	−10944
c_{17}	496315939418	c_{37}	13926575064245278	c_{57}	5700525398332	c_{77}	1378048
c_{18}	2566829630420	c_{38}	8111845655394708	c_{58}	−174467226040	c_{78}	−1964608
c_{19}	4960760844860	c_{39}	−2482436824724268	c_{59}	−1582514069956	c_{79}	500736
c_{20}	140805110136	c_{40}	−10166477180199681	c_{60}	−1430755264748	c_{80}	−36864

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 80, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $80 + 4$ terms removed returns order 80 as well, so $\deg q_{\min} = 80 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $80 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 80 that this sequence satisfies — and that family turns out to have exactly one member.

References

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